Machine safety specialist (CE)

Assignments - elaboration

TAB4: Risk reduction and standards
Tab4-Assignments 20231010

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Question 4-1

- **To which step does the E-STOP device belong??**

Machinery Directive:
- Emergency stop devices must be a back-up to other safeguarding measures and not a substitute for them.

EN IEC 60204-1:
- Emergency stop and emergency switching off are complementary protective measures that are not primary means of risk reduction for hazards (for example trapping, entanglement, electric shock or burn) at a machine

Answer:
Complementary protective measures, so, step 2 of the mandatory sequence

Exercise 4-2

- A company is developing a drill press as can be seen in the photo.
- To which steps of the imperative order do the following risk-reducing measures belong?:
  - Function-blocking shield (over v-belt)
    Step-2 protective measure
  - Safety control of the function-blocking protective cover
    Step-2 protective measure
  - "Safety glasses mandatory pictogram"”
    Step-3 Information for Use
  - Emergency stop device
    step-2 Additional protective measures
  - Operating instructions for the drill press
    Step-3 Information for Use

Question 4-3

What type of standards do the following belong to?

- EN 13854: Minimum distances to prevent entrapment? **Type B**
- EN 614-1: Ergonomic design principles**Type B**
- EN 1010-5: Corrugated board production machines and flat and corrugated board conversion machines**Type C**
- EN-IEC 62061: safety-related control systems (SIL) **Type B**
- EN-IEC 10218-1: Industrial robots**Type C**

Question 4-4

What tools are there to determine which standards apply to the risk assessment and reduction of a machine?

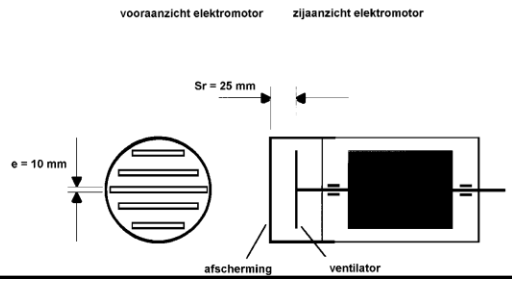
1. **List of harmonised standards**
2. **Search on NEN website, google and/or NEN Connect**
3. **Many references to other standards from NEN-EN-ISO 12100**

Exercise 4-5

EN-ISO 13857

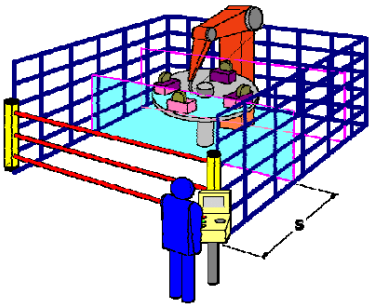
Reaching through a guard
See image and corresponding dimensions.

- Is the distance to the fan sufficient? **No, according to the standard it should be at least 80 mm.**
- If not, what are options to comply with the standard? **Increasing other shape openings (round/square) or distance to 80 mm.**



Exercise 4-6

- Machine stopping time is 0.3 sec.
- Determine the minimum safety distance.



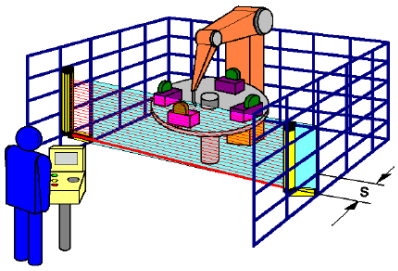
- $S = K \times T + C$

  $S = 1{,}6 \times 0{,}3 + 0{,}85$

  $S = 1{,}33\ m$

Exercise 4-7

- Machine stopping time is 0.3 s.
- Resolving power 14mm.
- Determine the minimum safety distance.



- With reaching in speed

  $S = 2 \times 0{,}3$

  $S = 0{,}60m \quad : S > 0{,}5\ m$
- With walking speed

  $S = 1{,}6 \times 0{,}3$

  $S = 0{,}48m \quad : S < 0{,}5\ m$ Choose $S = 0{,}50m$

Question 4-8

Which of the following risk-reducing measures are safety functions (SF) that can be developed with the PL/SIL standard?:

- *'Audible signal for the start of a dangerous movement'*
  No SF. Is carried out by the control but is not a 'step-2' measure but a step-3 measure (warning)

- *A guard around a dangerous part of the machine*
- **No SF. A 'step-2' measure but not because of the control system.**

- *Prevent a movement from starting as long as it detects that someone is present in the danger zone.* **A SF though. It is a 'step-2' measure which is carried out by the control system.**

- *Emergency stop function is* **A SF though. It is a 'step-2' measure which is carried out by the control system.**

Homework assignment module 4

1. Determine whether a type C standard is available for the case submitted.
2. Following the risk assessment (module 3), determine which risk mitigation measures should be applied, taking into account the imperative sequence.
3. Indicate which standards have been applied (list) and give some (total 3-5) examples of hazards and measures described herein and how they have been applied.

Submit next lesson.

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